# Personalized Differential Privacy in the Shuffle Model

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**Abstract.** Personalized local differential privacy is a privacy protection mechanism that aims to safeguard the privacy of data by using personalized approaches, while also providing practical data analysis results. It offers more flexible and precise privacy protection capabilities compared to traditional local differential privacy. By employing distinct privacy protection strategies for different users, it can better meet users' privacy requirements while minimizing the impact on data. However, existing mechanisms for personalized local differential privacy suffer from issues such as low query accuracy and poor data utility. These issues need to be addressed to improve the effectiveness and practicality of personalized local differential privacy.

In this work, we have proposed a framework of personalized differential privacy in the shuffle model. This framework introduces individualized perturbation to the data locally and then reshuffles the records in the dataset, disrupting the original order of the data and breaking the correlations between data points. This approach aims to achieve a higher level of privacy protection. We have validated the practicality and superiority of this framework on four different types of real-world datasets.

## 1 Introduction

With the rapid advancement of technology and the emergence of a data-driven economy, a vast amount of personal information is being collected and stored for various commercial purposes, such as personalized advertising, market research, and user behavior analysis. However, this widespread collection and utilization of data has also given rise to concerns regarding personal information leakage, identity theft, and infringement upon individual rights. Furthermore, significant events have further emphasized the need for privacy protection. For instance, the disclosure of the surveillance program conducted by the US National Security Agency, exposed by Edward Snowden, highlighted the government's capability to extensively monitor citizens' communications and personal data, triggering a debate on the delicate balance between privacy rights and national security. Similar incidents, coupled with data breaches, cyber attacks, and other related issues, have intensified people's focus on the security of personal information and the necessity for safeguarding privacy.

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In recent years, local differential privacy(LDP) [1,2] has emerged as a privacy protection mechanism based on individual privacy. It requires each participant to locally perturb or process data to enhance privacy. This approach allows for the protection of individual data privacy while enabling the server to analyze and utilize the uploaded data. Currently, local differential privacy has been applied in many real-world scenarios and has shown excellent performance. For example, Chrome by Google [3] and Windows 10 by Microsoft [4] use it for user data collection, and there have been numerous research outputs and explorations related to it in the scientific community. [5] applied LDP to frequency and mean estimation.

However, Local Differential Privacy (LDP) sacrifices data utility while protecting privacy. Hence, the concept of Shuffle model emerged. This mechanism shuffles the perturbed data with the help of a third party before it is sent to the server. The purpose of this operation is to disrupt the original order between the data and break the correlation between them, thereby achieving a higher level of privacy protection. The exist works such as Amplification-by-shuffling lemmas quantify how well the privacy parameters are improved [6,7], the single-message protocol [8,9] and m-message Shuffle Privacy [10,11]. But, the characteristic of shuffle model and LDP, which requires the same level of privacy protection for each individual, is not suitable for the majority of real-world scenarios. In real-world situations, due to different backgrounds of individuals, the privacy preferences for the same type of information vary among different users. For example, celebrities have much stricter privacy protection requirements for their flight information compared to ordinary people.

Based on the different privacy needs of individuals, the concept of personalized differential privacy(PDP) has emerged [12,13]. Personalized local differential privacy(PLDP) [12] allows for the development of different data perturbation strategies based on the characteristics and requirements of individual users. Compared to traditional shuffle model and LDP, personalized local differential privacy offers more flexible and precise privacy protection capabilities. By adopting different privacy protection strategies for different users, it is possible to better meet their privacy needs while minimizing the impact on data. This approach can to some extent improve data availability and the accuracy of data analysis. Individual users can choose the appropriate level of privacy protection based on their privacy preferences, thereby customizing more suitable personalized privacy protection solutions. [14] proposed a histogram optimization strategy based on personalized differential privacy. [15] presented a perturbation algorithm (PDPM) that satisfies personalized local differential privacy (PLDP), addressing the problem of inadequate or excessive privacy protection for certain participants caused by using the same privacy budget for all clients.

Although the PLDP model avoids the risk of storing all data centrally on a single server and enhances the security of data privacy, it has a significant gap in terms of data availability compared to the central differential privacy model. The main challenges of personalized local differential privacy currently are: (1) How to accurately model users' characteristics and privacy preferences

in order to generate appropriate personalized perturbations. (2) Balancing the level of privacy protection among different users, avoiding excessive protection or exposure of certain users' privacy. (3) Striking a balance between individual data privacy protection and data utility.

Based on the above challenges, in this paper, we propose several different privacy budget levels for users to choose from freely in order to meet the privacy preferences of different individuals and find the most suitable privacy protection strategy. We also separate the privacy budget parameters from the perturbed data during transmission to enhance the level of privacy protection. In addition, to further address the impact of local data perturbation on server-side data analysis, we have introduced the shuffle model. Finally, in the data aggregation phase on the server side, we assign fixed weights to each perturbed data for more accurate unbiased estimation. In this article, we investigated the frequency estimation problem in the context of personalized differential privacy in tne shuffle model. We improved the impact of noise perturbation on data availability by using a shuffling model and extended it to the multi-message shuffling model scenario. A large number of experimental results showed that our method significantly outperformed personalized local differential privacy.

In summary, we have made several contributions to this paper:

1. We have improved the basic framework of PLDP by separating the privacy parameters and perturbed data for enhanced privacy protection. Additionally, we have assigned fixed weights to each user to obtain more accurate unbiased estimates.
2. The first application of shuffle model in the PDP problem was introduced, and it was demonstrated that multi-message shuffling differential privacy method improve the utility of data statistics.
3. To demonstrate the effectiveness of our approach, we applied it to multiple real-world datasets of different types. The experimental results consistently showed that our method outperformed PLDP model, and we provided insights into the impact of certain parameters on utility.

## 2 Preliminaries

### 2.1 Differential Privacy

Differential Privacy(DP) is a privacy-preserving method that aims to minimize the risk of individual privacy disclosure during the analysis or processing of individual data. The core idea of differential privacy is to hide the specific information of individual data by introducing noise or randomness, allowing meaningful analysis of data while protecting privacy.

**Definition 1.** *Differential Privacy(DP) [16]: A random algorithm $\mathcal{R} : R(D) \rightarrow \mathcal{S}$ satisfies $(\epsilon, \delta)$-DP, where $\epsilon \geq 0$ and $0 \leq \delta \leq 1$, if and only if any Neighboring datasets $D, D'$, for any possible of outputs $z \subseteq \mathcal{S}$, it has*

$$Pr[\mathcal{R}(D) \in z] \leq e^{\epsilon} \cdot Pr[\mathcal{R}(D') \in z] + \delta \tag{1}$$

*where $\mathcal{S}$ represents the set of values output by $\mathcal{R}$. If $\delta = 0$, random algorithm $\mathcal{R}$ satisfies pure DP($\epsilon$-DP), if $\delta > 0$, $\mathcal{R}$ satisfies approximate DP($(\epsilon, \delta)$-DP).*

### 2.2 Shuffle Model

To address the issue of low utility in LDP models, some researchers have proposed Shuffle Differential Privacy(shuffle model) [6,8]. The shuffle model primarily provides privacy protection by introducing a semi-trusted third-party to reorganize or shuffle the perturbed data. The shuffle model serves as a compromise between central differential privacy models(CDP model) and LDP models. It neither relies on trusted servers nor requires excessive noise addition locally like the LDP model. Therefore, it can effectively enhance data utility in specific application scenarios.

**Definition 2.** *Shuffle model [17]: A shuffling model consists of three components: Randomizer, Shuffler and Analyzer.*

1. ***the Randomizer*** *$\mathcal{R}$: $\mathcal{R}(a, \cdot) \rightarrow y^m$: The randomizer $\mathcal{R}$ perturbs the input real data $a$ and output $y^m$. When $m = 1$, it is referred to as a single-message protocol, and when $m > 1$, it is referred to as a multi-message protocol.*
2. ***the Shuffler*** *$\mathcal{S}$: $\mathcal{S}(y^m) \rightarrow y^m$: By recombining the attribute values of records or exchanging them with other records, the shuffler makes the correlation and individual information of the dataset more obscure and indistinguishable.*
3. ***the Analyzer*** *$\mathcal{A}$: $A(y^m) \rightarrow \mathcal{Z}$: The analyzer is responsible for collecting and processing the messages, as well as conducting subsequent analysis tasks.*

*If we use protocol $\mathcal{P}(\mathcal{R} \circ \mathcal{S} \circ \mathcal{A})$ to describe the above process, we define this process as:*

$$\mathcal{P}(a) = \mathcal{A}(\mathcal{S}(\cup_{i=1}^{n} \mathcal{R}_i(a_i, \cdot)) \tag{2}$$

Based on the above content, we can derive the following definition 3.

**Definition 3.** *Differential Privacy in shuffle model(shuffle DP). the protocol $\mathcal{P}$ satisfies $(\epsilon, \delta)$-DP, where $\epsilon \geq 0$ and $0 \leq \delta \leq 1$, if and only if $\mathcal{S}(\cup_{i=1}^{n} \mathcal{R}_i(a_i, \cdot))$ satisfies central $(\epsilon, \delta)$-DP where $\mathcal{S}(\cdot)$ is the shuffle operation.*

**Definition 4.** *Hockey-stick divergence. The hockey-stick divergence between two random variables $P$ and $Q$ is defined by:*

$$\mathbf{D}_{e^\epsilon}(P \| Q) = \int_{max} \{0, P(x) - e^\epsilon Q(x)\} \, dx \tag{3}$$

*iff $max\{\mathbf{D}_{e^\epsilon}(P \| Q), \mathbf{D}_{e^\epsilon}(Q \| P)\} < \delta$ that $P$ and $Q$ are $(\epsilon, \delta)$-indistinguishable.*

### 2.3 Personalized Differential Privacy

Personalized Differential Privacy(PDP) is an extended form of differential privacy that aims to provide more precise privacy protection while maintaining the availability and utility of individual data. Personalized differential privacy allows data processors to customize privacy protection mechanisms for each user based on their privacy requirements and data characteristics.

**Definition 5.** *Personalized Differential Privacy [12]. Given the local user sets* $\mathcal{U} = \{u_1, u_2, \ldots, u_n\}$ *and* $\varepsilon = \{\epsilon_1, \epsilon_2, ..., \epsilon_n\}$*, where each user* $u_i \in \mathcal{U}$ *has a privacy preference* $\epsilon_i \in \varepsilon$*, and a random mechanism* $R : R(D) \to \mathcal{S}$ *satisfying* $(\varepsilon, \delta)$*-PDP where* $0 \leq \delta \leq 1$*,*$0 \leq \epsilon_i \leq 1$ *. if and only if any Neighboring datasets* $D, D'$*, for any possible of outputs* $z \subseteq \mathcal{S}$*, it has*

$$Pr[\mathcal{R}(D) \in z] \leq e^{\epsilon_i} \cdot Pr[\mathcal{R}(D') \in z] + \delta \tag{4}$$

*where* $\mathcal{S}$ *represents the set of values output by* $\mathcal{R}$ *and* $\epsilon_i$ *is the user* $u_i$*'s privacy preference.*

## 3 System Overview

### 3.1 Problem Definition

In a distributed scenario, data is typically stored across different nodes or participants, and it may involve multiple privacy-sensitive data owners. Frequency estimation can potentially leak information about individuals. Even without directly disclosing sensitive data, attackers may be able to infer individual attributes or behavioral patterns through analyzing frequency estimation results. Therefore, privacy concerns are an important consideration. In the case of an untrusted server, we typically employ the PLDP mechanism to provide personalized privacy protection for local users.

We use $\varepsilon = \cup_{i=1}^{n} \epsilon_i$ to define the set of privacy budget values, where $\epsilon_i$ represents the privacy preference of local user $u_i$. Smaller values of $\epsilon_i$ indicate stronger privacy. Suppose there is a local user population where each user holds an element $a_i \in \mathcal{T}$, where $\mathcal{T} = \{t_1, t_2, t_3, ..., t_k\}$. And let

$$f_t = \frac{1}{n} \sum_{i=1}^{n} 1_{(a_i = t)} \tag{5}$$

$f_t$ denotes the frequency of a specific element $t$ in the entire population. $n$ represents the total number of users. This method is commonly used in fields such as analyzing user behavior, conducting statistical surveys, and data mining.

Although PLDP improves certain limitations of the application scenarios, such as the single-purpose nature and limited privacy protection strength of LDP mechanisms, there still exists a significant gap in terms of accuracy compared to the CDP. Our goal is to maximize the statistical utility of server-side data while achieving personalized privacy protection.

### 3.2 Muti-messages Personalized Differential Privacy in Shuffle Model

We achieve this goal by introducing the shuffle model. Shuffle model is similar to LDP that perturbs individual data locally. However, unlike LDP, shuffle model enhances privacy protection by introducing a third-party shuffler that

anonymizes and randomly reorders the perturbed data. Therefore, when both mechanisms provide the same level of privacy protection, the shuffle DP allows for the addition of less noise locally, resulting in a significant improvement in statistical utility at the server-side.

In this paper, we have introduced a scenario of Muti-messages personalized differential privacy in shuffle model(Muti-messages PSDP). We specify that in addition to the privacy budget $\epsilon_i$, each user can send $m$ messages, among which only one is a true message, and the rest are false messages sampled from a random uniform distribution. All messages are then sent to a random shuffler for shuffling before being sent to the server for data aggregation(see Fig. 1).

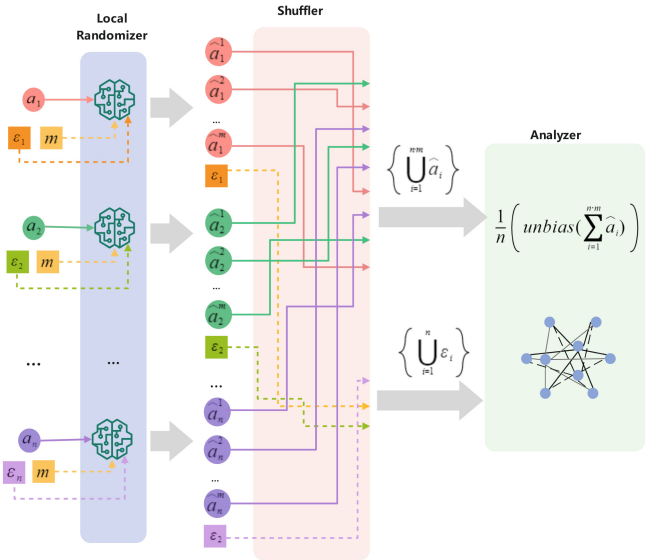


**Fig. 1.** The framework of multi-PSDP

**The Local Randomization Perturbation(Randomizer).** The local randomization perturbation (Randomizer) and server-side aggregation analysis(Analyzer). The local Randomizer perturbs the original data by modifying it with a certain probability or adding random noise to increase the indistinguishability between two pieces of data. The most commonly used mechanism is the random response mechanism [18] and its generalized versions [3]. Generalized Randomized Response (GRR) [19,20] is more suitable for situations where $k > 2$. Therefore, we have chosen GRR as the local randomizer to perturb the real data. The workflow is as follows:

$$\mathbf{P}\left[\mathcal{R}(a) = a^*\right] = \begin{cases} \frac{e^{\epsilon}}{e^{\epsilon}+k-1}, a^* = a \\ \frac{1}{e^{\epsilon}+k-1}, a^* \neq a \end{cases} \tag{6}$$

The local users $u_i$ input their data $a_i$ and the selected privacy budget $\epsilon_i$ into the local randomizer. The randomizer perturbs the real data and uniformly sample $m-1$ false messages from $\mathcal{T}$, then send the perturbed $m$ messages and the privacy budget $\epsilon_i$ to the shuffler. The specific details are described in Algorithm 1.

**Algorithm 1:** the local random perturbation mechanism

**Input**: $a_i$: a ture value of local user
$\mathcal{T}$: the dataset of total values
$m$:The number of messages sent by local users.
**Parameter**: $\epsilon_i$: the local privacy budget satisfy shuffle $(\varepsilon, \delta)$-PDP
$k$: the cardinality of total values

$\rho = \frac{e^{\epsilon_i}-1}{e^{\epsilon_i}+k-1}$
$y = Ber(\rho)$
**if** $y = 1$ **then**
  $\hat{a}_i = a_i$
**end if**
**else**
  choose $\hat{a}_i$ from $\mathcal{T}$ uniformly
**end if**
$\mathbb{S}_i \leftarrow \mathbb{S}_i \uplus \hat{a}_i$
$\rho = (m-1) - \lfloor m-1 \rfloor$
$y = Ber(\rho) + \lfloor m-1 \rfloor$
**for** $j \leftarrow 1$ *to* $y$ **do**
  choose $\hat{a}_i^j$ from $\mathcal{T}$ uniformly
  $\mathbb{S}_i \leftarrow \mathbb{S}_i \uplus \hat{a}_i^j$
**endfor**
**Send** $\mathbb{S}_i, \epsilon_i$ **to shuffler separately**

**Third-Party Shuffler.** At this stage, Shuffler shuffles the perturb values dataset $\mathbb{S}$ and privacy budget dataset $\varepsilon$ to remove the correlation between data and send them to server.

**Data Aggregation and Estimation(Analyzer).** In the frequency estimation problem, the server needs to aggregate the received data and perform unbiased estimation to leverage the statistical utility of the data. he specific details are described in Algorithm 2. Since the correspondence between messages and privacy budgets is disrupted after being randomly shuffled, we simplify the data aggregation and unbiased estimation calculations by assigning a fixed weight $w_i$ to each user.

The process of aggregation and unbiased estimation is as follows:

$$\begin{aligned}\hat{f}_t &= \frac{1}{W}(\sum_{i=1}^{n}(\frac{\hat{a}_i - \frac{m-1}{k} - q_i}{p_i - q_i} \cdot w_i)) \\ &= \frac{1}{W}(\hat{N}_t - n \cdot \frac{m-1}{k} - \sum_{i=1}^{n} q_i)\end{aligned} \tag{7}$$

We set $w_i = p_i - q_i$, $W = \sum_{i=1}^{n} w_i$ and $\hat{N}_t$ be the total number of received value $t$ after aggregation, the estimated frequency of value $t$ is $\hat{f}_t$.

**Algorithm 2:** Server-side data aggregation and estimation

**Input**: $\mathbb{S}$: the perturb dataset
$\varepsilon$: privacy budget dataset
**Parameter**: $N$: total numbers of user
$k$: the cardinality of total values
**Output**: $\hat{f}_t$

**for** $\epsilon_i \in \varepsilon$ **do**
  $p_i = \frac{e^{\epsilon_i}}{e^{\epsilon_i}+k-1}$
  $q_i = \frac{1}{e^{\epsilon_i}+k-1}$
  $W = W + (p_i - q_i)$
**endfor**
get $\hat{N}_t = \sum_{i=1}^{n \cdot m} 1_{\hat{s}_i = t}$ from any data $\hat{s}_i \in \mathbb{S}$
$\hat{f}_t = \frac{1}{W}(\hat{N}_t - n \cdot \frac{m-1}{k} - \sum_{i=1}^{n} q_i)$
**Return** $\hat{f}_t$

### 3.3 Privacy Analysis

To analyze the privacy of the recommended method, we introduce the concept of personalized differential privacy in shuffle model and prove the privacy amplification effect of our method.

According to Definitions 3, 2 and 5, we have formalized the definition of personalized differential privacy in shffle model.

**Definition 6.** *Personalized differential privacy in shuffle model(shuffle-PDP): Given the sets $\mathcal{U} = \{u_1, u_2, \ldots, u_n\}$, and $\varepsilon = \{\epsilon_1, \epsilon_2, ..., \epsilon_n\}$, where each user $u_i \in \mathcal{U}$ has a privacy preference $\epsilon_i \in \varepsilon$ and a value $a_i \in D$, and a shuffle protocal $\mathcal{P} : \mathcal{R} \circ \mathcal{S}(D) \to \mathcal{T}$ satisfying $\varepsilon$-PDP ,if and only if any Neighboring datasets $D, D'$. Then the algorithm :*

$$\mathcal{R} \circ \mathcal{S}(\cdot) = \mathcal{S}(R_1(a_1, \epsilon_1, m), R_2(a_2, \epsilon_2, m), ..., R_n(a_n, \epsilon_n, m)) \tag{8}$$

*is satisfies $(\varepsilon, \delta)$- shuffle PDP, where the $\mathcal{T}$ is the output dataset, $m$ represents the number of messages sent by each user.*

Here, to further enhance the privacy amplification effect, we employ the framework of the Variation-ratio [21], a unified, efficient, and comprehensive framework, to analyze privacy amplification in shuffling models. Therefore, we are given two neighboring datasets $\mathcal{A}$ and $\mathcal{A}'$. i.e. $\mathcal{A} = \left\{a_1 = a_1^0, a_2, a_3..., a_n\right\}$, $\mathcal{A}' = \left\{a_1 = a_1^1, a_2, a_3..., a_n\right\}$ where $\mathcal{A}, \mathcal{A}' \subset \mathbb{A}$.

These two datasets differ only in the first data entry, we defined some parameters $p > 1, \beta \in [0, \frac{p-1}{p+1}], q > 1$ on local randomzer $\{\mathcal{R}_i\}_{i\in[n]}$:

1. $(p, \beta)$-variation property: for all possible $a_1^0, a_1^1 \in \mathbb{A}$ that satifies the formula $\mathbf{D}_p(\mathcal{R}_1(a_1^0)\|\mathcal{R}_1(a_1^1)) = 0$ and $\mathbf{D}_{e^0}(\mathcal{R}_1(a_1^0)\|\mathcal{R}_1(a_1^1)) \leq \beta$.
2. $q$-ratio property:for all possible $a_1, a_2, a_3..., a_n \in \mathbb{A}$ and all $\{\mathcal{R}_i\}_{i\in[2:n]}$ that satifies $\mathbf{D}_q(\mathcal{R}_1(a_1)\|\mathcal{R}_i(a_i)) = 0$ .

**Lemma 1.** *(Variation-ratio reduction [21]) For* $p > 1, \beta \in [0, \frac{p-1}{p+1}], q > 1$*,let* $C \sim Bin(n-1, \frac{2\beta p}{(p-1)q}), A \sim Bin(C, \frac{1}{2})$ *and* $\Delta_1 \sim Ber(\frac{\beta p}{p-1})$ *and* $\Delta_2 \sim Ber(1 - \Delta_1, \frac{\beta}{p-1-\beta p})$*;let* $P^q_{\beta,p}$ *denote* $(A + \delta_1, C - A + \Delta_2)$ *and* $Q^q_{\beta,p}$ *denote* $(A + \Delta_2, C - A + \Delta_1)$*. Given any* $a_1, a_2, a_3..., a_n \in \mathbb{A}$*,if* $\{\mathcal{R}_i\}_{i\in[n]}$ *satisfy the* $(p, \beta)$*-variation property and the* $q$*-ratio property, then for any measurement* $D$ *satisfying the data-processing inequality:*

$$D(S(\mathcal{R}_1(a_1^0), ..., \mathcal{R}_n(a_n))\|S(\mathcal{R}_1(a_i^0), ..., \mathcal{R}_n(a_n)) \leq D(P^q_{\beta,p}\|Q^q_{\beta,p}) \tag{9}$$

Now, we are attempting to prove that our method satisfies Definition 6 within the framework of the Variation-ratio.

According to the GRR mechanism used in our method, we convert the parameters $p', \beta', q'$ in the framework of the Variation-ratio, where the privacy preference $\epsilon \in \varepsilon$ .The process is as follows:

First, according to the Lemma 1, given the randomizer $\mathcal{R}$ satisfy the $(p', \beta')$-property and $q'$-ratio property that:

$$\mathbf{D}_{p'}(\mathcal{R}(a_1^0)\|\mathcal{R}(a_1^1)) = 0$$

The application of Definition 4 can transform the above equation into:

$$\int_{max} \left\{0, \mathcal{R}(a_1^0) - p' \cdot \mathcal{R}(a_1^1)\right\} da = 0$$

therefore:

$$\mathcal{R}(a_1^0) - p' \cdot \mathcal{R}(a_1^1) = 0$$

so,we can get a $p'$ satisfy the above conditions when the GRR mechanism is used in $\mathcal{R}$:

$$p' = \frac{\mathcal{R}(a_1^0)}{\mathcal{R}(a_1^1)} = \frac{\frac{e^\epsilon}{e^\epsilon + k - 1}}{\frac{1}{e^\epsilon + k - 1}} = e^\epsilon$$

Similarly, we can obtain a suitable $\beta'$ according to the $(p', \beta')$-variation property:

$$\mathbf{D}(\mathcal{R}(a_1^0)\|\mathcal{R}(a_1^1) \leq \beta'$$

The application of Definition 4 can transform the above equation into:

$$\int_{max} \left\{0, \mathcal{R}(a_1^0) - \mathcal{R}(a_1^1)\right\} da \leq \beta'$$

so,we can get the $\beta'$ be bound by $[0, \frac{e^\epsilon - 1}{e^\epsilon + k - 1}]$.

And in multi-message scenario, we get false messages from $\mathcal{T}$ uniformly. Similarly, we get the parameter $q'$ when $\mathcal{R}_{i \in [2:n]}$ satisfy $q'$-ratio property as follow:

$$\mathbf{D}_{q'}(\mathcal{R}(a_1) \| \mathcal{R}_i(a_i)) = 0$$

Similarity,wo can get the $q'$ which satisfy the above conditions that: $q' = \frac{k \cdot e^\epsilon}{e^\epsilon + k - 1}$

if there are $n'$ users and each user contributes $m-1$ fake messages, the value of $n-1$ in Defintion 1 becomes $n' \cdot (m-1)$. Therefore, by applying Definition 1, our method satisfy:

$$D(\mathcal{R} \circ \mathcal{S}(R_1(a_1^0, \epsilon_1, m), \cup_{i=2}^n \mathcal{R}_i(a_i, \epsilon_i, m)) \| R \circ \mathcal{S}(R_1(a_1^1, \epsilon_1, m), \cup_{i=2}^n \mathcal{R}_i(a_i, \epsilon_i, m)) \leq D(P_{\beta', p'}^{q'} \| Q_{\beta', p'}^{q'})$$

are $(\epsilon_1, \delta)$-indistinguishable according to Definiton 4, where $\epsilon_1 \in \varepsilon$. So, our method satisfy $(\varepsilon, \delta)$-shuffle PDP in framwork of the Variation-ratio.

## 4 Experiments

In this chapter, we evaluate and compare the performance of our proposed methods, single-message PSDP and multi-message PSDP, through extensive experiments. Since there is no existing work specifically addressing frequency estimation in the PSDP scenario, we compare our work with frequency estimation in the PLDP scenario. We conduct tests on various types of real data to demonstrate the impact of personalized perturbation and different hyperparameter settings on performance through experiments.

### 4.1 Datasets

Four commonly used datasets will be used in our experiments. These datasets are widely used for training and testing frequency estimation problems. The specific information about the datasets is provided in Table 1. The Adult dataset [22], which was extracted by Barry Becker from the 1994 Census database, contains 48,842 instances. For our dataset, we extracted a subset of 32561 samples from it. Bank Marketing [23] is related to a direct marketing campaign conducted by a Portuguese banking institution. We extracted a subset of 10,000 samples from this dataset to create a new dataset. Abalone [24] is derived from physical measurements of abalone (a type of sea snail). Car Evaluation [25] contains several basic parameters of cars and is derived from a simple hierarchical decision model.

**Table 1.** Dataset

| Datasets | Feature | Sample numbers | dimension |
|---|---|---|---|
| Adult | education | 32561 | 16 |
| Bank marketing | job | 10000 | 12 |
| Abalone | sex | 4177 | 3 |
| Car evaluation | buy | 1728 | 4 |

### 4.2 Performance Metrics

In this study, we use the Mean Squared Error (MSE) evaluation method to assess performance. The MSE is a commonly used evaluation metric to measure the degree of difference between predicted values and true values. It calculates the squared difference between predicted values and true values, and then takes the average. A smaller MSE indicates a smaller difference between the predicted results and the true values, indicating a better performance of the model. MSE is widely used in many machine learning and statistical fields.

$$MSE = \sum_{i=1}^{k} (f_t - \hat{f}_t)^2 \tag{10}$$

### 4.3 Experimental Performance

In this section, we compare our method with PLDP and single-PSDP on four real datasets under two different scenarios of privacy distributions (see Table 2). We further investigate the impact of message quantity and privacy distribution on experimental performance and present the results using line graphs for better visualization.

**Table 2.** Experimental parameter settings

| | Scenario 1 | Scenario 2 |
|---|---|---|
| $\epsilon$ allocation | 0.2,0.4,0.6,0.8,1.0 | 0.3,0.5,1.0 |
| User ratio | 0.25,0.25,0.25,0.25,0.25 | 0.33,0.33,0.34 |

PLDP allows users to freely choose the level of privacy protection according to their preferences, providing flexibility to adapt to various complex privacy-security scenarios. Although personalized local differential privacy partially addresses the limitations of LDP mechanisms, such as limited applicability and limited privacy protection strength, there is still a significant gap in accuracy compared to the CDP model. As shown in the Fig. 2, regardless of the scenario,

the performance of the PLDP method is the worst. Both single-PSDP and local differential privacy perturb the local individual data items. However, unlike local differential privacy, single-PSDP further enhances privacy protection by introducing a third-party shuffler to anonymize and randomly reorder the perturbed data. When the privacy protection strengths of both mechanisms are the same, the single-PSDP mechanism allows for less noise to be added locally. Therefore, as shown in the graph, the results of single-PSDP have improved significantly compared to PLDP, but there is still a considerable gap compared to Our method.

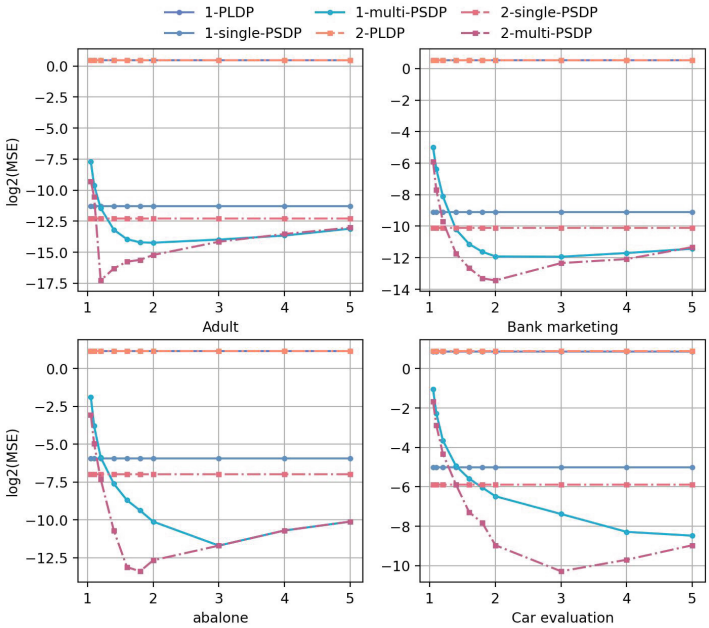


**Fig. 2.** Comparison of experimental results

**Impact of the Dataset.** We tested our method using four datasets of varying sizes and dimensions. Regardless of whether it was in scenario 1 or scenario 2, our method performed the best on the Adult dataset and the worst on the Car Evaluation dataset. The results indicate that with an increase in the number of samples, the experimental performance significantly improves. It can be observed that our method performs better and demonstrates its advantages more effectively on large-scale datasets.

**Impact of Message Quantity.** We explored the impact of the quantity of fake messages on the experimental performance. Although increasing the quantity of fake messages can amplify the privacy budget disturbance of the GRR mechanism on real data, excessive fake news still introduces a significant amount of noise to the aggregated results. Therefore, we set several reasonable message numbers(msg) to find the optimal parameter for the quantity of messages. In theory, as the quantity of messages increases, the MSE value initially decreases before increasing, with the rate of decrease gradually slowing down. From the Fig. 2, it can be observed that the optimal number of messages are as follows: Adult: msg = 2 (Scenario 1), msg = 1.4 (Scenario 2). Bankmarketing: msg=3 (Scenario 1), msg = 2 (Scenario 2). Analone: msg = 3 (Scenario 1), msg = 1.8 (Scenario 2). Car evaluation: msg = 5 (Scenario 1), msg = 3 (Scenario 2). Taking all factors into consideration, a reasonable range for the message number in $[1.4, 3]$.

**Impact of Privacy Parameters $\epsilon$.** We have also explored the impact of privacy partition granularity on experimental performance. Generally, a finer granularity of privacy preference partitioning can flexibly address more complex privacy and security scenarios but may have an impact on experimental performance. In Fig. 2, in all datasets, in the single-PSDP method, the experimental results for Scenario 2 consistently outperform Scenario 1. In the multi-PSDP method, the experimental results for Scenario 2 are generally better than Scenario 1 within a certain range, and gradually converge as the number of messages increases.

## 5 Future Work

This work focuses on personalized differential privacy. To strike a balance between privacy and utility, we propose a personalized shuffling differential privacy framework for multi-message scenarios. Under this framework, we evaluate multiple different types of real datasets, and the results show that our approach outperforms existing methods signif- icantly, effectively improving centralized privacy protection. In the future, our work will focus on the following aspects of expansion: (1) Maintaining utility in more complex privacy protection scenarios for data analysis. (2) Exploring more efficient coding techniques or data sparsification tech- niques to balance communication costs and statistical utility.

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